📌 **Problem Statement**

Telecom Customer Churn Analysis

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1. Business Background

The telecom industry is highly competitive, and customer churn — the loss of customers over time — is a significant challenge. High churn rates negatively impact revenue, increase customer acquisition costs, and reduce overall profitability. Analyzing historical customer data helps telecom companies understand churn behaviour and monitor key customer segments more effectively.

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2. Business Problem

Although the telecom company collects detailed customer-level data, it lacks structured analytical insights into churn behaviour. This limits the company's ability to understand which customers are churning, how churn varies across customer groups, and how usage behaviour differs between churned and retained customers.

Key questions addressed:

- What is the overall customer churn rate?
- Which customer segments show a higher likelihood of churn?
- How do usage patterns (calls, SMS, data usage) differ between churned and retained customers?
- How does churn vary across telecom partners and demographic groups?

Goal:
To use SQL-based exploratory data analysis and Power BI visualizations to examine churn patterns and enable stakeholders to monitor churn trends and customer segments effectively.

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3. Dataset Information

The dataset contains customer-level records, where each row represents an individual telecom customer.

Category	Attributes
Identification	Customer ID, date of registration
Demographics	Age, gender
Geography	State, city, pincode *(excluded from analysis due to reliability issues)*
Telecom Partner	Telecom partner name
Usage Metrics	Calls made, SMS sent, data used

Category	Attributes
Financial Indicator	Estimated salary
Churn Label	Binary indicator (0 = retained, 1 = churned)

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4. Business Objectives

The objectives of this analysis are to:

- Measure overall churn metrics, including total customers, churned customers, and churn rate
- Identify high-risk customer segments based on age, gender, and telecom partner
- Analyse customer usage behaviour (calls, SMS, data usage) in relation to churn
- Examine tenure patterns using customer registration dates to understand churn timing
- Explore estimated salary differences between churned and retained customers as a proxy for customer value
- Build interactive Power BI dashboards to support churn monitoring and exploratory analysis for business stakeholders

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5. Tools and Technologies

Purpose	Tool / Technology
Data extraction & querying	SQL (MySQL)
Exploratory Data Analysis (EDA)	SQL
Data visualization & dashboards	Power BI

Power BI is used to create interactive dashboards with KPIs, filters, and visual comparisons that allow stakeholders to explore churn trends without writing queries.

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6. Key Metrics & Analysis

The analysis focuses on the following metrics:

- **Churn Rate:** Percentage of customers who have churned
- **Customer Count Metrics:** Total customers vs churned customers
- **Segmented Churn Analysis:** Churn by age group, gender, and telecom partner
- **Usage Behaviour Analysis:** Comparison of calls, SMS usage, and data consumption between churned and retained customers
- **Tenure Analysis:** Churn patterns based on registration date

- **Financial Proxy Analysis:** Comparison of estimated salary across churn groups

Note: Average Revenue Per User (ARPU) is **not calculated**, as the dataset does not contain actual revenue or billing information.

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7. Power BI Dashboard Features

The Power BI dashboard includes:

- **Summary KPIs:** Total customers, churned customers, churn rate
- **Demographic Visuals:** Churn by age group and gender
- **Telecom Partner Analysis:** Churn distribution by telecom partner
- **Usage Comparison Charts:** Calls, SMS, and data usage by churn status
- **Trend Visuals:** Churn patterns over time based on registration date
- **Interactive Filters:** Age, gender, telecom partner, churn status

These dashboards enable interactive exploration of churn patterns and customer behaviour.

✅ **One-Line Summary**

This project analyses telecom customer churn using SQL and Power BI to explore churn patterns across demographics, usage behaviour, and telecom partners, while excluding unreliable geographic fields from analysis.